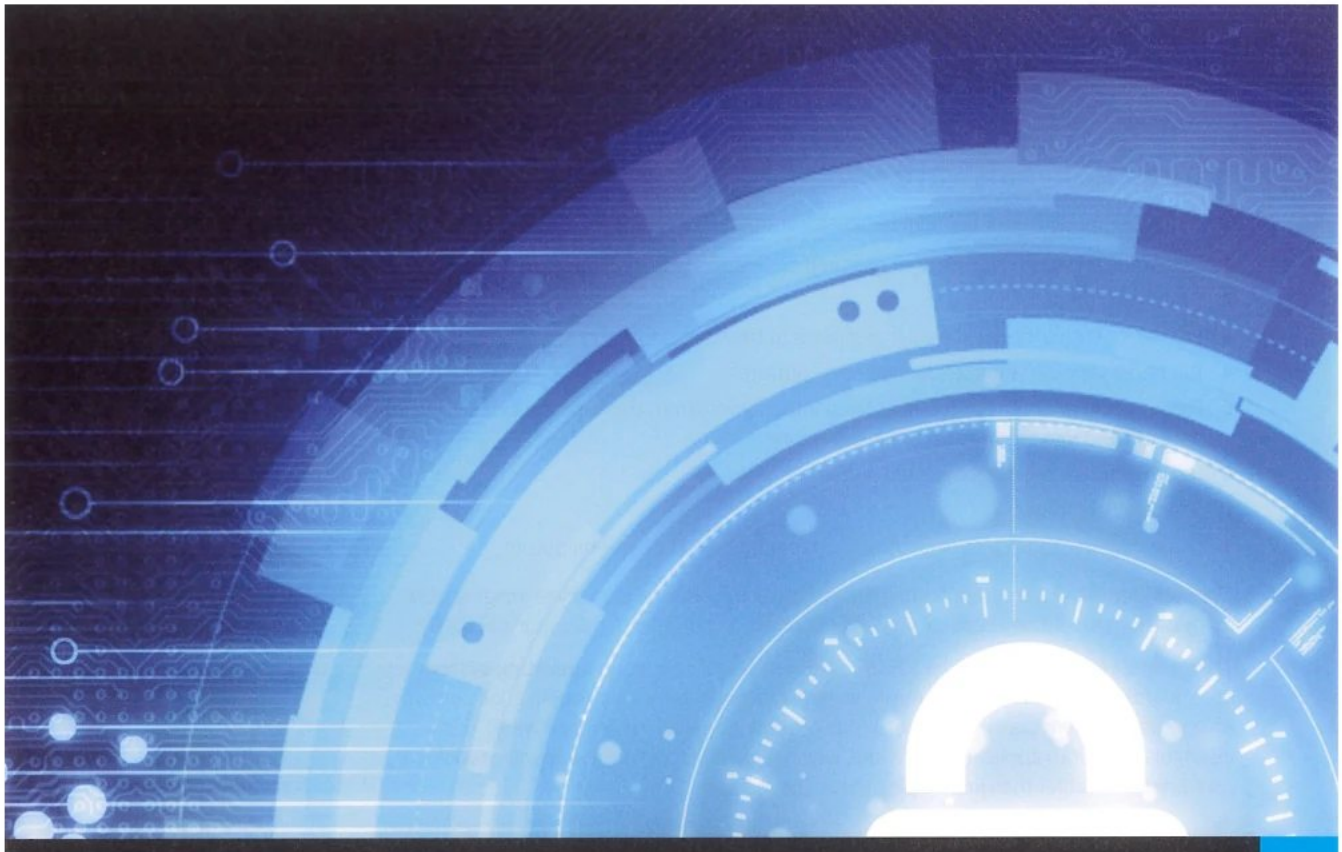


Chapter 8:
System software

Learning objectives

By the end of this chapter you should be able to:

- explain why a computer system requires an Operating System (OS)
- explain the key management tasks carried out by the Operating System
- show understanding of the need for typical utility software provided with an Operating System
- show understanding of program libraries
- show understanding of the need for:
  - assembler software for the translation of an assembly language program
  - a compiler for the translation of a high-level language program
  - an interpreter for translation and execution of a high-level language program
- explain the benefits and drawbacks of using either a compiler or an interpreter and justify the use of each
- show awareness that high-level language programs may be partially compiled and partially interpreted, such as Java
- describe features found in a typical Integrated Development Environment (IDE).

8.01 System software

In the 1960s, the likely arrangement for using a computer would be something like this.

1. Enter machine room with deck of punched cards and a punched paper tape reel.
2. Switch on computer.
3. Put deck of cards into card reader and press button.
4. Put paper tape into tape reader and press button.
5. Press button to run the program entered into memory from the punched cards using the data entered into memory from the paper tape.
6. Press button to get output printed on the line-printer.
7. Switch off computer.
8. Leave machine room with deck of cards, paper tape and line-printer output.

The user controlled the computer hardware by pressing buttons. Just try to imagine how many buttons would be needed if you had to control a computer in the same way today.

The missing component from the 1960s computer was, of course, an **operating system**; in other words, some software to control the hardware and interact with application software. An operating system is an example of a type of software called 'system software'. This distinguishes it from application software, which is created to perform a specific task for a computer user rather than just helping to run the system.

KEY TERM

Operating system: a software platform that provides facilities for programs to be run which are of benefit to a user

8.02 Operating system activities

Operating systems are extremely complex and it is not possible to give a full description here of what an operating system is. However, what an operating system generally does is to provide an environment where programs can be run that are of benefit to a user.

The activities of an operating system can be sub-divided into different categories, some of which overlap with each other. We are going to look at each of the various tasks carried out by the operating system. Details of how some of them are carried out are discussed in Chapter 20 (Sections 20.01 to 20.05).

User–system interface

A user interface is needed to allow the user to get the software and hardware to do something useful. An operating system should provide at least the following for user input and output:

- a command-line interface
- a graphical user interface (GUI).

Discussion Point:

Have you any experience of using a command-line interface?

Program–hardware interface

Programmers write software and users run this software. The software uses the hardware. The operating system has to ensure that the hardware does what the software wants it to do. Program development tools associated with a programming language allow a programmer to write a program without needing to know the details of how the hardware, particularly the processor, actually works. The operating system then has to provide the mechanism for running the developed program.

Resource management

When a program has started to run it is described as a **process**. In a modern computer system, a process will not be able to run to completion without interruption. At any time there will be many processes running on the computer system. Each process needs access to the resources provided by the computer system.



KEY TERM

Process: a program that has begun execution

The resource management provided by the operating system aims to achieve the best possible efficiency in computer system use. The two most important aspects of this are:

- scheduling of processes
- resolution of conflicts when two processes require the same resource.

Memory management

There are three important aspects of memory management.

- Memory protection ensures that one program does not try to use the same memory locations as another program.
- The memory organisation scheme is chosen to achieve the best usage of limited memory size, for example, virtual memory involving paging or segmentation.
- Memory usage optimisation involves decisions about which processes should be in main memory at any one time and where they are stored in this memory.

Device management

Every computer system has a variety of components that are categorised as 'devices'. Examples include the monitor screen, the keyboard, the printer and the webcam. The management of these requires:

- installation of the appropriate device driver software
- control of usage by processes.

File management

Three major features here are the provision of:

- file naming conventions
- directory (folder) structures
- access control mechanisms.

Security management

Chapter 9 (Sections 9.02, 9.03 & 9.04) and Chapter 21 contain extensive accounts of security issues and measures used to prevent problems. For the operating system the many aspects of security management include:

- provision for recovery when data is lost
- prevention of intrusion
- ensuring data privacy.

Error detection and recovery

Errors can arise in the execution of a program either because it was badly written or because it has been supplied with inappropriate data. Other errors are associated with devices not working correctly. Whatever the cause of an error, the operating system should have the capability to interrupt a running process and provide error diagnostics where appropriate. In extreme cases, the operating system needs to be able to shut down the system in an organised fashion without loss of data.

TASK 8.01

For each of the above categories of operating system task, each point could be placed in a different category. Make an abbreviated list of these categories and add arrows to show different categories where each point could be placed.

Question 8.01

It is useful to describe the management tasks carried out by an operating system as being primarily one of the following types:

- those assisting the user of the system
- those concerned with the running of the system.

Considering the management tasks that have already been categorised, can you identify them as belonging to one or other of the above types? Are there any problems in doing this?

8.03 Utility software

A utility program can be provided by the operating system or it can be installed separately. It is a program that is not executed as part of the normal running of the operating system. Instead it is a program that the user or the operating system can decide to run when needed. For example, some utility programs manage hard disks.

Hard disk formatter and checker

A disk formatter will typically carry out the following tasks:

- removing existing data from a disk that has been used previously
- setting up the file system on the disk, based on a table of contents that allows a file recognised by the operating system to be associated with a specific physical part of the disk
- partitioning the disk into logical drives if this is required.